Reading Statistics in Papers

LET Mathematics & Statistics

Example from Sackley et al. (2016)

This workshop uses data from the research paper: “An occupational therapy intervention for residents with stroke related disabilities in UK care homes (OTCH): cluster randomised controlled trial” by Sackley et al.

Table 1: Details of clusters and personal and baseline assessment information for participants. Values are numbers (percentages) unless stated otherwise.

Characteristics	Intervention group	Control group
Care home type		
Residential care	53/114 (46)	54/114 (47)
Nursing care	61/114 (54)	60/114 (53)
Mean (SD) cluster size	5 (3.7)	4.2 (3.0)
Personal details		
Mean (SD) age (years)	83.1 (9.9)	83.6 (9.5)
Men	203/568 (36)	174/474 (37)
White	517/568 (91)	445/474 (94)
Comorbidities		
Cardiovascular disease	342/530 (65)	278/446 (62)
Fall history	203/495 (41)	200/427 (47)

Consider Table 1

1. Look at the values in brackets across Table 1. Do they always represent the same thing?
2. What was the mean age for the participants in the control group?
3. What was the standard deviation of age for the participants in the control group?
4. What percentage of the intervention group had cardiovascular disease?
5. What percentage of the intervention group had **not** had a fall history?

6. From the descriptive statistics in Table 1 alone, do the two groups seem similar at baseline?

Table 2: Comparison of primary and secondary outcomes at six and 12 month follow-up assessment

Outcome by follow-up	Intervention Adj. Mean (SE)	Intervention (No)	Control Adj. Mean (SE)	Control (No)	Difference in adj. means (95% CI)	P value
Barthel index‡						
6 months	4.78 (0.20)	525	4.78 (0.22)	448	0.004 (−0.52 to 0.53)	0.99
12 months	3.93 (0.21)	512	3.77 (0.22)	430	0.16 (−0.40 to 0.72)	0.58
Rivermead mobility index						
6 months	2.64 (0.11)	421	2.67 (0.12)	346	−0.03 (−0.33 to 0.27)	0.84
12 months	2.19 (0.13)	354	2.46 (0.14)	271	−0.26 (−0.62 to 0.09)	0.15
Geriatric depression scale-15						
6 months	6.20 (0.21)	338	6.68 (0.22)	284	−0.48 (−1.04 to 0.09)	0.10
12 months	6.22 (0.22)	297	6.40 (0.25)	219	−0.18 (−0.80 to 0.43)	0.56
EQ-5D-3L§						
6 months	0.22 (0.02)	363	0.23 (0.02)	315	−0.01 (−0.05 to 0.04)	0.72
12 months	0.20 (0.02)	316	0.18 (0.02)	244	0.02 (−0.03 to 0.07)	0.48

*Adjusted by care home as a random effect, and baseline score, type of care home and centre as fixed effects.
†Tukey-Kramer adjusted confidence intervals and P values.
‡Participants who died before follow-up are given a Barthel score of zero.
§EuroQol group 5-dimension self report questionnaire (three levels).

Consider Table 2 and the **Barthel Index**:

7. For the Barthel index after 6 months, what is the adjusted mean for the control group and intervention group?

8. What does the number in the bracket after the mean represent?

   ○ Standard Deviation

   ○ Standard Error

   ○ Percentage

9. What was the difference in adjusted means between the intervention and control groups at 6 months?

10. What was the 95% confidence interval for the difference in adjusted means at 6 months?

11. Was the value of '0' (representing no difference) within the 95% confidence interval for the difference in adjusted means?

12. Is there any evidence of a difference in the Barthel index between the control and intervention groups at 6 months?

13. Do any of the 6-month or 12-month comparisons (Barthel, Mobility, Depression, EQ-5D) show evidence of a difference?

14. Look at footnote ‡. Why did the researchers give participants who died a Barthel score of zero?

15. The difference for the Rivermead mobility index at 12 months is -0.26. Based on how these differences are usually calculated (Intervention minus Control), what does the negative sign indicate?

16. The Barthel index is a 20-point scale. If a study found a difference between groups that was statistically significant ($p < 0.05$) but only 0.004 points, how should a clinician react?

    ○ Adopt the intervention immediately based on the p-value

    ○ Recognize that while statistically significant, the effect size is likely too small to be clinically meaningful to a patient

    ○ Ignore the result because p-values are invalid when the mean difference is small.